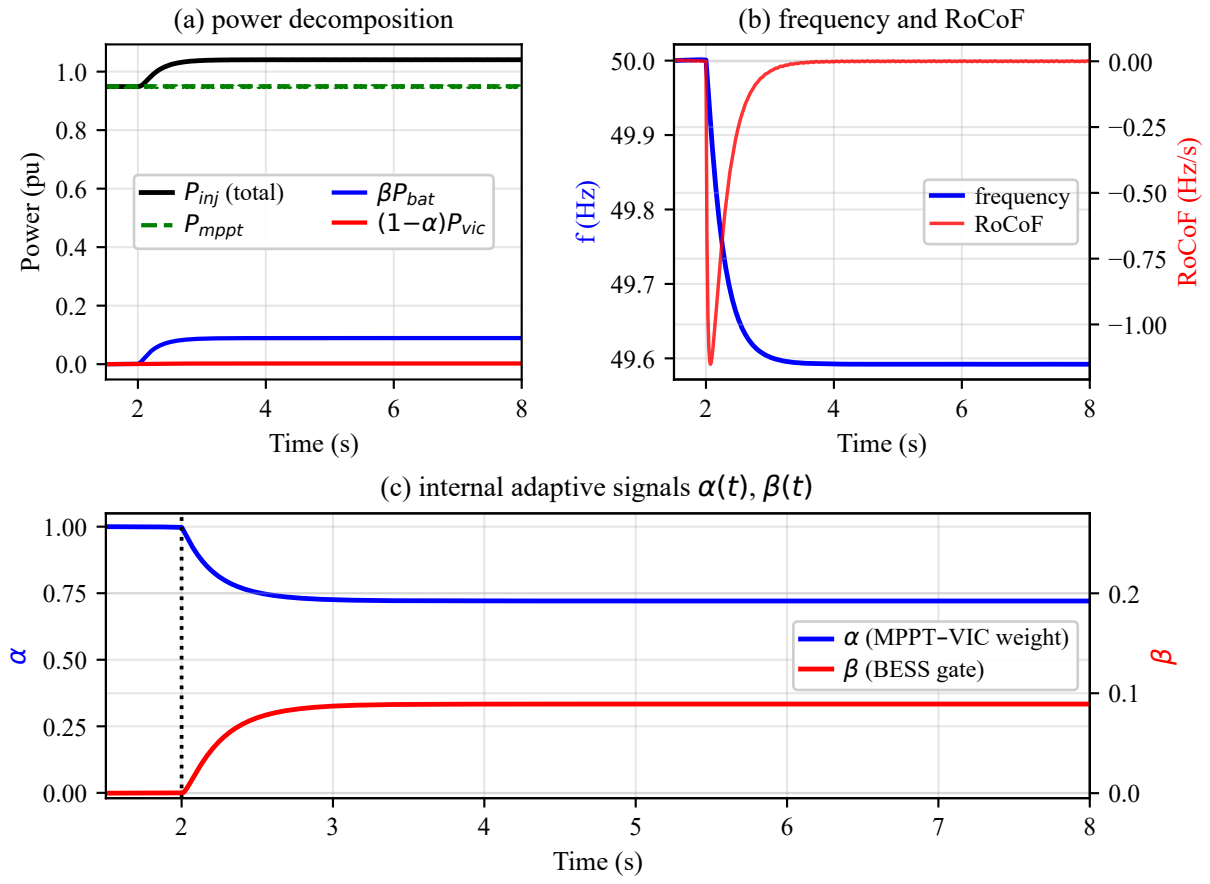

(a) power decomposition
Power (pu)
Time (s)
P_{inj} (total)
P_{mppt}
βP_{bat}
$(1-\alpha)P_{vic}$
(b) frequency and RoCoF
f (Hz)
RoCoF (Hz/s)
Time (s)
frequency
RoCoF
(c) internal adaptive signals $\alpha(t)$, $\beta(t)$
α
β
Time (s)
α (MPPT–VIC weight)
β (BESS gate)